图示电路为380 / 220 伏三相四线制供电系统, 接有两组对称三相负载, 已知：$Z = 40 + j30\Omega$ (其余参数如图所示).

试求：

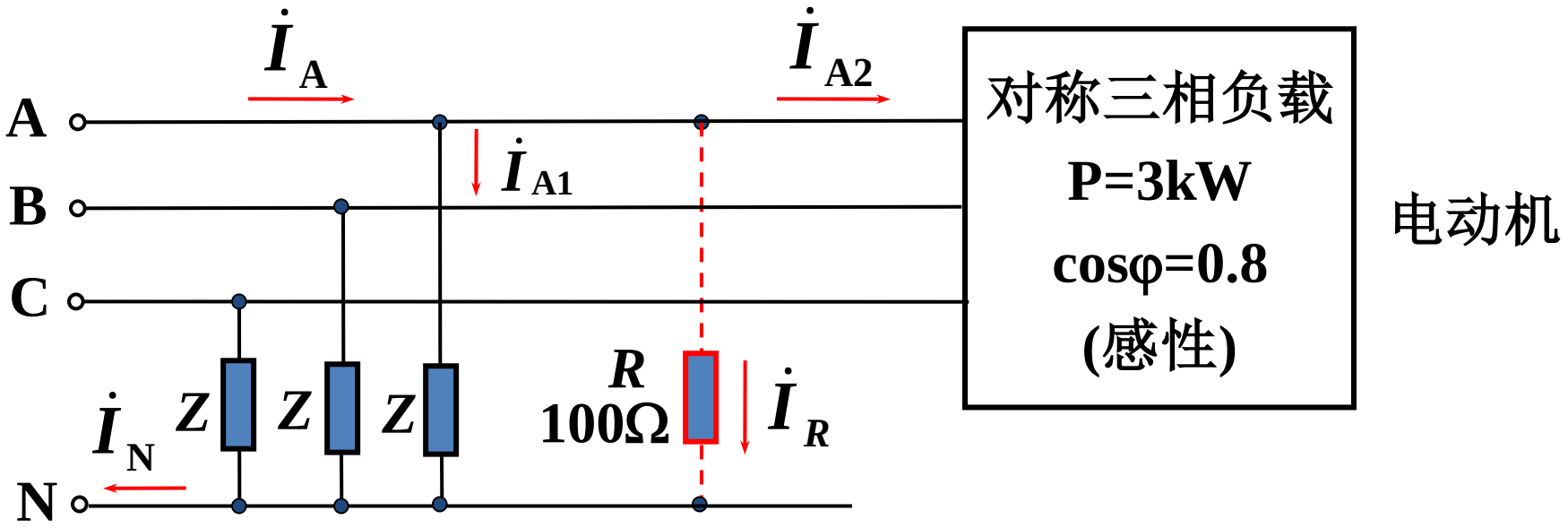
(1) 三个线电流 $\dot{I}_A$, $\dot{I}_{A1}$, $\dot{I}_{A2}$和中线电流$\dot{I}_N$;

(2) 计算三相电源发出的总功率;

(3) 画出两表法测电动机功率的接线图, 计算功率表读数;

(4) 若Z为可选择的电抗，则当Z = ? 时，三相总功率因数为1;

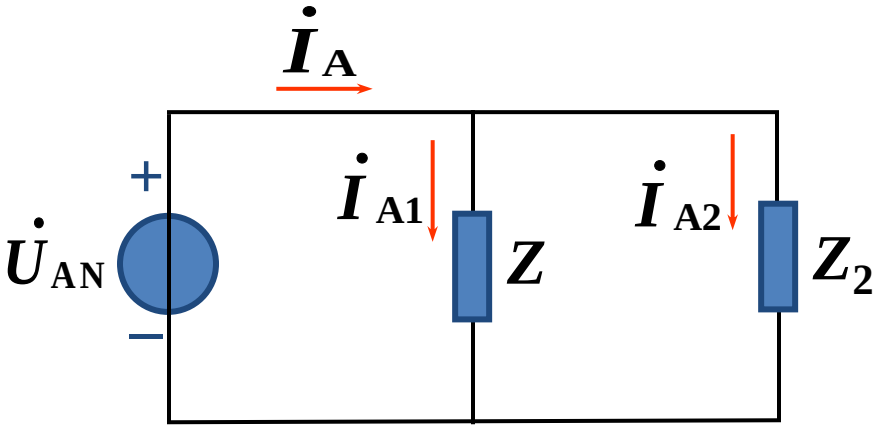
(5) 在A相接一100Ω 电阻, 重答(1). 若此时断开中线求 R 中电流.



解： 令 $\dot{U}_{AN}=220\angle 0^\circ \text{V}$

(1) 设第二组对称三相负载为Y接，每相负载阻抗为Z_2，

则一相计算电路为：



解法一：直接用功率求I_{A2}

$$\dot{I}_{A1}=\frac{\dot{U}_{AN}}{Z}=\frac{220\angle 0^\circ}{40+\text{j}10}$$

$$=\frac{220\angle 0^\circ}{41.23\angle 14.04^\circ}=5.336\angle -14.04^\circ \text{A}$$

$$I_{A2}=\frac{P}{\sqrt{3}U_l\cos\varphi}=\frac{3000}{\sqrt{3}\times 380\times 0.8}=5.698\text{A}$$

由 $\cos\varphi=0.8$，得

$\varphi=36.87^\circ$ (感性) $\therefore \dot{I}_{A2}=5.698\angle -36.87^\circ \text{A}$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} = 5.336\angle -14.04^\circ + 5.698\angle -36.87^\circ$$

$$= (5.117 - j1.295) + (4.557 - j3.421)$$

$$= 9.735 - j4.714 = 10.82\angle -25.84^\circ \text{ A}$$

其它两相的总电流：

$$\dot{I}_B = 10.82\angle -145.8^\circ \text{ A}, \quad \dot{I}_C = 10.82\angle 94.16^\circ \text{ A}$$

由对称性，得 $\dot{I}_N = 0$

解法二：求出Z_2

$$I_{A2} = \frac{P}{\sqrt{3}U_l \cos\varphi} = \frac{3000}{\sqrt{3}\times 380\times 0.8} = 5.698\text{A}$$

$$|Z_2| = \frac{U_{AN}}{I_{A2}} = 38.61\Omega$$

$\varphi = 36.87^\circ$ (感性)　　$Z_2 = |Z_2|\angle\varphi = 38.61\angle 36.87^\circ\Omega$

$$Z_{总}=\frac{ZZ_2}{Z+Z_2}=\frac{41.23\angle 14.04^\circ\times 38.61\angle 36.87^\circ}{41.23\angle 14.04^\circ+38.61\angle 36.87^\circ}$$

$$=\frac{1591.9\angle 50.94^\circ}{70.88+\mathrm{j}33.18}=\frac{1591.9\angle 50.94^\circ}{78.26\angle 25.10^\circ}=20.34\angle 25.84^\circ\Omega$$

$$\dot{I}_{\mathrm{A}}=\frac{\dot{U}_{\mathrm{AN}}}{Z_{总}}=\frac{220\angle 0^\circ}{20.34\angle 25.84^\circ}=10.82\angle -25.84^\circ\,\mathrm{A}$$

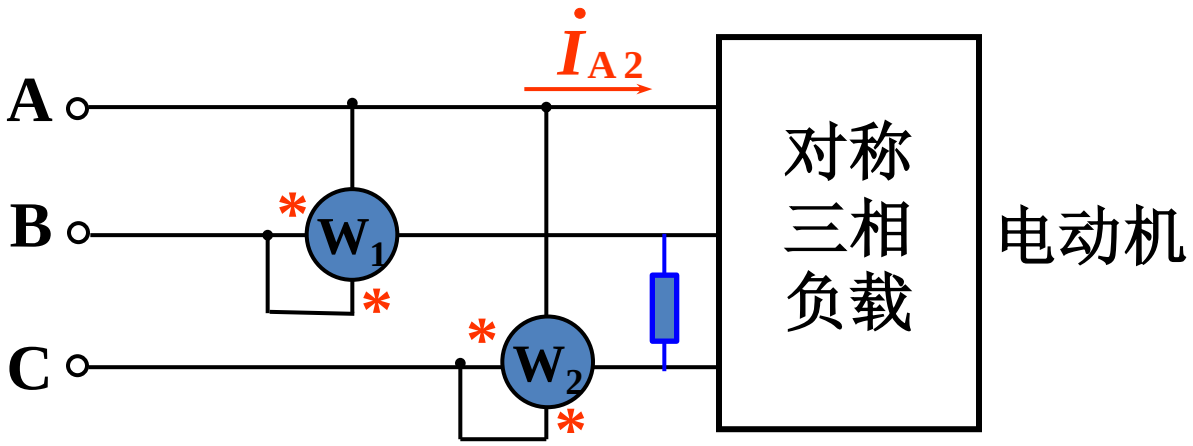
(2) 三相电源发出的总功率

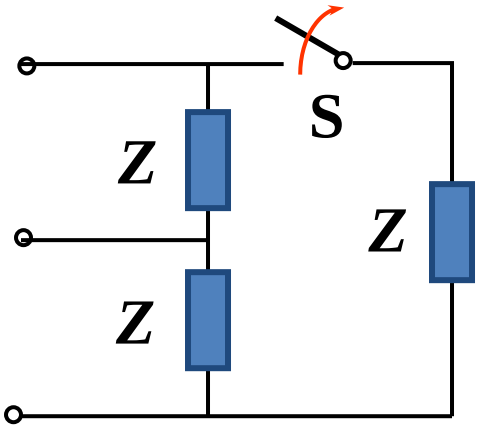
$$P=\sqrt{3}U_l I_{\mathrm{A}}\cos\varphi'=\sqrt{3}\times 380\times 10.82\times\cos 25.84^\circ=6.41\mathrm{kW}$$

或

$$P=P_1+P_2=\sqrt{3}\times 380\times 5.336\times\cos 14.04^\circ+3000=6.41\mathrm{kW}$$

(3) 两表法测电动机负载功率的接线如图。





功率表W_1、 W_2的读数分别为P_{21}、 P_{22}:

$$P_{21} = U_{BA} I_B \cos\varphi'$$

$$= 380 \times 5.698 \times \cos[(-150^\circ - (-156.87^\circ)] = 2.150\text{kW}$$

$$P_{22} = U_{CA} I_C \cos\varphi''$$

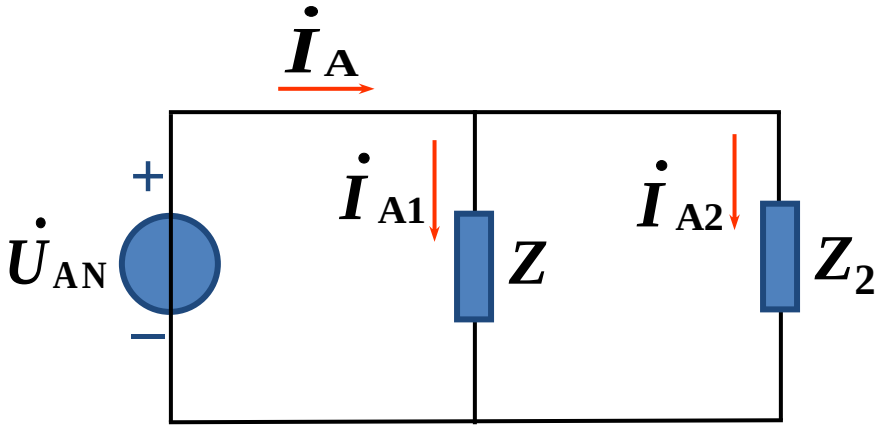
$$= 380 \times 5.698 \times \cos[(150^\circ - 83.13^\circ)] = 849.5\text{W}$$

$$P_2 = P_{21} + P_{22} \approx 3000\text{W}$$

(4) 由第（1）问解法二已经算出电动机等效阻抗为：

$$Z_2=|Z_2|\angle\varphi = 38.61\angle 36.87°\Omega$$

则Z= -jX$_C$



$$\text{虚部}\left\{\frac{1}{-jX_c}+\frac{1}{38.61\angle 36.87°}\right\}=0$$

$$\frac{1}{X_c}+\frac{1}{38.61}\sin\left(-36.87°\right)=0$$

$$X_c=\frac{38.61}{\sin\left(36.87°\right)}$$

(5) 接入电阻R后，当有中线时，原三相负载电流不变。

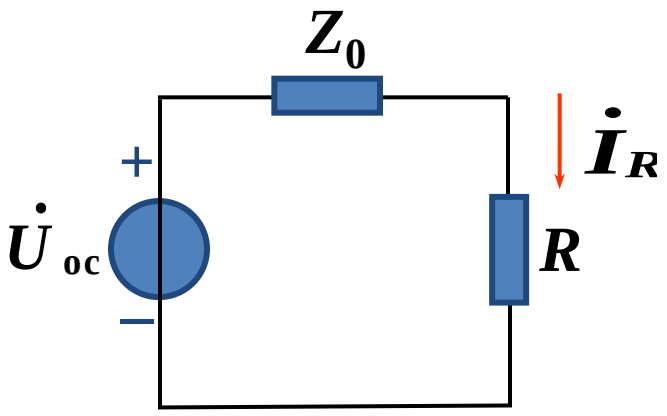
$$\dot{I}_R = \frac{\dot{U}_{AN}}{R} = \frac{220\angle 0^\circ}{100} = 2.2\angle 0^\circ \text{ A} \qquad \dot{I}_N = \dot{I}_R = 2.2\angle 0^\circ \text{ A}$$

$$\dot{I}_A = \dot{I}_{A1} + \dot{I}_{A2} + \dot{I}_R = 5.336\angle -14.04^\circ + 5.698\angle -36.87^\circ + 2.2\angle 0^\circ$$

$$= 9.735 - j4.714 + 2.2 = 911.93 - j4.714 = 12.83\angle -21.6^\circ \text{ A}$$

$\dot{I}_B$，$\dot{I}_C$不变.

当中线断开，可由戴维南定理得等效电路：



$$Z_0 = Z/3 = 13.33 + j3.33\Omega$$

$$\dot{U}_{oc} = \dot{U}_{AN} = 220\angle 0^\circ \text{ V}$$

$$\dot{I}_R = \frac{\dot{U}_{oc}}{Z_0 + R} = \frac{220\angle 0^\circ}{13.33 + j3.33 + 100}$$

$$= \frac{220\angle 0^\circ}{133.4\angle 1.43^\circ} = 1.65\angle -1.43^\circ \text{ A}$$